

Give me 20 common study quiz questions for Data Structures

Arrays

1. What is an array, and what are its advantages and disadvantages?
2. What is the time complexity of inserting an element at the beginning, middle, and end of an array?

Linked Lists

3. What is the difference between a singly linked list, doubly linked list, and circular linked list?
4. Why is insertion and deletion faster in a linked list than in an array?

Stacks

5. What is a stack, and which principle does it follow?
6. What are some real-world applications of stacks?

Queues

7. What is the difference between a queue and a stack?
8. What is a circular queue, and why is it used?

Trees

9. What is a binary tree?
10. What is the difference between a binary tree and a binary search tree (BST)?
11. What are the preorder, inorder, and postorder traversals?
12. What is the height of a tree?

Heaps

13. What is a heap, and what is the difference between a min-heap and a max-heap?

Hashing

14. What is a hash table?

15. What is a hash collision, and how can it be resolved?

Graphs

16. What is the difference between BFS (Breadth-First Search) and DFS (Depth-First Search)?

17. What is the difference between directed and undirected graphs?

18. What are weighted and unweighted graphs?

Algorithms & Complexity

19. What is Big O notation? Explain the time complexities of $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, and $O(n^2)$ with examples.

20. What is the difference between recursion and iteration? When would you prefer one over the other?